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Control apparatus, lens apparatus, image pickup apparatus, camera system, and control method

Abstract

A control apparatus for a camera system that includes a lens apparatus including a first optical system and a second optical system and configured to move them relative to each other, and an image pickup apparatus including an image sensor and detachably attachable to the lens apparatus includes a processor configured to acquire a first evaluation value of the first optical system at a first focus detection position corresponding to the first optical system, determine a second focus detection position corresponding to the second optical system based on an first object image formed by the first optical system and a second object image formed by the second optical system, acquire a second evaluation value of the second optical system at the second focus detection position, and move the first optical system and the second optical system based on the first evaluation value and the second evaluation value.

Inventors: **Nitto; Kiyoshi (Saitama, JP), Fukai; Yosuke (Kanagawa, JP), Murakami; Taro (Tokyo, JP), Fujiwara; Yusuke (Tokyo, JP)**

Applicant: **CANON KABUSHIKI KAISHA** (Tokyo, JP)

Family ID: **90574034**

Assignee: **CANON KABUSHIKI KAISHA (Tokyo, JP)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2006/0033990	12/2005	Kato	359/425	G02B 7/06
2009/0146045	12/2008	Yamaguchi	250/201.4	H04N 23/13
2016/0295097	12/2015	Shanmugavadivelu	N/A	H04N 23/45
2024/0137637	12/2023	Ebisawa	N/A	H04N 23/80

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2009-175498	12/2008	JP	N/A

Primary Examiner: Nazrul; Shahbaz

Attorney, Agent or Firm: Cowan, Liebowitz & Latman, P.C.

Background/Summary

BACKGROUND

Technical Field

(1) One of the aspects of the embodiments relates to a control apparatus, a lens apparatus, an image pickup apparatus, a camera system, and a control method.

Description of Related Art

(2) A lens apparatus has conventionally been known in which a pair of left and right optical systems are disposed apart from each other by a predetermined distance (base length), and two image circles are imaged in parallel on a single image sensor. For this lens apparatus, images formed by the pair of left and right optical systems are recorded as moving images or still images for the left eye and the right eye, respectively, and when viewed using a three-dimensional display, VR goggles, etc. during playback, the right eye of the viewer views the image for the right eye and

his left eye views the image for the left eye. At this time, the images with parallax are projected to the right and left eyes due to the base length of the pair of left and right optical systems, so the viewer can acquire a stereoscopic effect.
(3) The pair of left and right optical systems to capture images with parallax needs focusing for each of the pair of left and right optical systems.
(4) Japanese Patent Laid-Open No. 2009-175498 discloses binoculars that uses a single operation member that switches between left and right diopter adjustment by moving one optical system and focusing by moving both optical systems.
(5) The binoculars disclosed in Japanese Patent Laid-Open No. 2009-175498 performs the diopter adjustment and focusing by switching between them with a single operation member, and the operation becomes complicated and proper focusing is difficult due to erroneous operation.
SUMMARY
(6) A control apparatus according to one aspect of the embodiment for a camera system that includes a lens apparatus including a first optical system and a second optical system and configured to move the first optical system and the second optical system relative to each other, and an image pickup apparatus including an image sensor and attachable to and detachable from the lens apparatus includes a memory storing instructions, and a processor configured to execute the instructions to acquire a first evaluation value of the first optical system at a first focus detection position corresponding to the first optical system, determine a second focus detection position corresponding to the second optical system based on an first object image formed by the first optical system and a second object image formed by the second optical system, acquire a second evaluation value of the second optical system at the second focus detection position, and move the first optical system and the second optical system based on the first evaluation value and the second evaluation value.
(7) A control apparatus according to another aspect of the embodiment for a camera system that includes a lens apparatus including a first optical system and a second optical system and is configured to move the first optical system and the second optical system relative to each other, and an image pickup apparatus that includes an image sensor and is attachable to and detachable from the lens apparatus includes a memory storing instructions; and a processor configured to execute the instructions to acquire a first evaluation value of the first optical system at a first focus detection position corresponding to the first optical system, and move the first optical system and the second optical system based on the first evaluation value.
(8) Each of a lens apparatus, an image pickup apparatus, and a camera system including the above control apparatus also constitutes another aspect of the embodiment.
(9) Further features of the disclosure will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a sectional view of an interchangeable lens according to one embodiment.
(2) FIGS. **2**A and **2**B are exploded perspective views of an interchangeable lens.
(3) FIG. **3** illustrates a positional relationship between each optical axis and image circles on an image sensor.
(4) FIG. **4** is a schematic configuration diagram of a camera system according to one embodiment.
(5) FIG. **5** is a block diagram of the interchangeable lens according to Example 1.
(6) FIG. **6**A is an example of a flowchart illustrating a focusing operation of a camera system according to Example 1.
(7) FIG. **6**B is another example of a flowchart illustrating a focusing operation of the camera

system according to Example 1.
(8) FIG. **7** is a flowchart illustrating an operation of the camera system with the interchangeable lens according to Example 1 attached.
(9) FIG. **8** is a flowchart illustrating a method of bundle adjustment according to Example 1.
(10) FIG. **9** is a schematic configuration diagram of a camera system according to Example 2.
(11) FIG. **10** is a schematic configuration diagram illustrating a tilted image sensor.
(12) FIGS. **11**A, **11**B, and **11**C illustrate an example of a focusing operation procedure according to Example 2.
(13) FIGS. **12**A and **12**B illustrate another example of the focusing operation procedure according to Example 2.
(14) FIGS. **13**A and **13**B illustrate another example of the focusing operation procedure of Example 2.
(15) FIGS. **14**A and **14**B illustrate other examples of focusing operation procedures according to Examples 2 and 3.
(16) FIG. **15** is a schematic configuration diagram of a camera system according to Example 5.
(17) FIGS. **16**A and **16**B illustrate a focusing operation procedure according to Example 5.
(18) FIG. **17** is a schematic configuration diagram of a camera system according to Example 6.
(19) FIG. **18** is a bottom view of an interchangeable lens according to Example 7 viewed from a lens mount side.
(20) FIG. **19** is a top view of the interchangeable lens according to Example 7 viewed from an object side.
(21) FIG. **20** is a top view of the interchangeable lens viewed from the object side, with the first optical system and the second optical system added to the configuration of FIG. **19** according to Example 7.
(22) FIGS. **21**A and **21**B are schematic configuration diagrams of the camera system according to Example 4.

DESCRIPTION OF THE EMBODIMENTS

(23) In the following, the term “unit” may refer to a software context, a hardware context, or a combination of software and hardware contexts. In the software context, the term “unit” refers to a functionality, an application, a software module, a function, a routine, a set of instructions, or a program that can be executed by a programmable processor such as a microprocessor, a central processing unit (CPU), or a specially designed programmable device or controller. A memory contains instructions or programs that, when executed by the CPU, cause the CPU to perform operations corresponding to units or functions. In the hardware context, the term “unit” refers to a hardware element, a circuit, an assembly, a physical structure, a system, a module, or a subsystem. Depending on the specific embodiment, the term “unit” may include mechanical, optical, or electrical components, or any combination of them. The term “unit” may include active (e.g., transistors) or passive (e.g., capacitor) components. The term “unit” may include semiconductor devices having a substrate and other layers of materials having various concentrations of conductivity. It may include a CPU or a programmable processor that can execute a program stored in a memory to perform specified functions. The term “unit” may include logic elements (e.g., AND, OR) implemented by transistor circuits or any other switching circuits. In the combination of software and hardware contexts, the term “unit” or “circuit” refers to any combination of the software and hardware contexts as described above. In addition, the term “element,” “assembly,” “component,” or “device” may also refer to “circuit” with or without integration with packaging materials.
(24) Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the disclosure. Corresponding elements in respective figures will be designated by the same reference numerals, and a duplicate description thereof will be omitted.
(25) A camera system according to one embodiment includes a lens apparatus (interchangeable

lens) that includes two optical systems (first optical system and second optical system) disposed in parallel (symmetrically), and an image pickup apparatus that configured to image two image circles in parallel on a single image sensor. The two optical systems are horizontally arranged, and separated by a base length. Viewed from the image side, an image formed by the right optical system (first optical system) is recorded as a moving or still image for the right eye, and an image formed by the left optical system (second optical system) is recorded as a moving image or still image for the left eye. By viewing a moving image or a still image (video) using a three-dimensional display, so-called VR goggles, or the like, the viewer's right eye views the right-eye image, and his left eye sees the left-eye image. At this time, images with parallax are projected to the right and left eyes due to the base length of the left and right optical systems, so the viewer can acquire a three-dimensional effect. Thus, the camera system according to this embodiment is a camera system for stereoscopic imaging that can form two images with parallax by the first optical system and the second optical system.

(26) FIG. **1** is a sectional view of an interchangeable lens **200** according to one embodiment. FIGS. **2**A and **2**B are exploded perspective views of the interchangeable lens **200**. In the following description, descriptions of the first optical system (right-eye optical system) are denoted by R, and descriptions of the second optical system (left-eye optical system) are denoted by L. Descriptions that are common to both the right-eye optical system and the left-eye optical system do not have the R or L suffix. Although the optical systems are disposed on the left and right sides in this embodiment, they may be disposed on the upper and lower sides.

(27) The interchangeable lens **200** includes a first optical system **201**R and a second optical system **201**L. Each of the two optical systems is fixed to a lens top base **300** with a screw or the like, and is capable of imaging with an angle of view of 120 degrees or more. Each of the two optical systems has, in order from the object side, a first optical axis OA**1**, a second optical axis OA**2** substantially orthogonal to the first optical axis OA**1**, and a third optical axis OA**3** parallel to the first optical axis OA**1**. Each of the two optical systems includes a first lens (unit) **211** having a convex surface **211**A on the object side disposed along the first optical axis OA**1**, a second lens (unit) **221** disposed along the second optical axis OA**2**, third lenses (lens units) **231**A and **231**B disposed along the third optical axis OA**3**. Each of the two optical systems further includes a first prism **220** that bends a light beam parallel to the first optical axis OA**1** and guides it to the second optical axis OA**2**, and a second prism **230** that bends a light beam parallel to the second optical axis OA**2** to the third optical axis OA**3**. In the following description, the optical axis direction is a direction parallel to the first optical axis OA**1**, which is the direction extending to the object side and the imaging surface side.

(28) FIG. **3** illustrates a positional relationship between each optical axis and image circles on the image sensor. A right-eye image circle ICR with an effective angle of view formed by the first optical system **201**R and a left-eye image circle ICL with an effective angle of view formed by the second optical system **201**L are arranged in parallel on an image sensor **111** of a camera body **110**. A size OA**2**, of each image circle and a distance between the image circle may be set so that the image circles do not overlap each other. For example, the light receiving range of the image sensor **111** is divided into left and right halves with respect to the center, The center of the right-eye image circle ICR may be set to an approximate center of the right area of the light receiving range, and the center of the left-eye image circle ICL may be set to an approximate center of the left area of the light receiving range.

(29) In this embodiment, each optical system is a full-circumference fisheye lens, and an image formed on the imaging surface is a circular image covering a range of angle of view exceeding 180 degrees, as illustrated in FIG. **3**, two circular images are formed on the left and right sides. A distance between the first optical axis OA**1**R of the first optical system **201**R and the first optical axis OA**1**L of the second optical system **201**L is referred to as base length L**1**. The longer the base length L**1** is, the more the stereoscopic effect becomes during viewing. For example, assume that a

sensor size is 24 mm long×36 mm wide, a diameter of the image circle is Φ17 mm, a distance L**2** between the third optical axes is 18 mm, and the length of the second optical axis is 21 mm. In a case where each optical system is arranged such that the second optical axis extends in the horizontal direction, the base length L**1** is 60 mm, which is approximately equal to the interpupillary distance of an adult. The diameter **1**D of the lens mount unit **202** may be shorter than the base length L**1**, and by making the distance L**2** between the third optical axes shorter than the diameter ΦD of the lens mount unit **202**, the three lens units **231**A and **231**B can be disposed inside the lens mount unit **202**. That is, a relationship of L**1**>ΦD>L**2** is established.

(30) In viewing as a VR, the angle of view that gives a three-dimensional effect is about 120 degrees, but since a 120-degree field of view leaves a sense of discomfort, the angle of view is often increased up to 180 degrees. In this embodiment, the effective angle of view exceeds 180 degrees, and the size ΦD**3** of the image circle in the range of 180 degrees is smaller than the size ΦD**2** of the image circle.

(31) FIG. **4** is a schematic configuration diagram of the camera system **100**. The camera system **100** includes an interchangeable lens **200** and a camera body **110** to which the interchangeable lens **200** is detachably attached.

(32) The interchangeable lens **200** includes the first optical system **201**R, the second optical system **201**L, and a lens system control unit **209**. The camera body **110** includes the image sensor **111**, an A/D converter **112**, an image processing unit **113**, a display unit **114**, an operation unit **115**, a memory **116**, a body system control unit **117**, and a camera mount unit **122**.

(33) In a case where the interchangeable lens **200** is attached to the camera body **110** via the lens mount unit **202** and the camera mount unit **122**, the body system control unit **117** and the lens system control unit **209** are electrically connected.

(34) A right-eye image formed via the first optical system **201**R and a left-eye image formed via the second optical system **201**L are formed side by side on the image sensor **111** as object images. The image sensor **111** converts each formed object image (optical signal) into an analog electrical signal. The A/D converter **112** converts the analog electrical signal output from the image sensor **111** into a digital electrical signal (image signal). The image processing unit **113** performs various image processing on the digital electrical signal output from the A/D converter **112**.

(35) The display unit **114** displays various information. The display unit **114** is realized by using an electronic viewfinder or a liquid crystal panel, for example. The operation unit **115** functions as a user interface for a photographer to give instructions to the camera system **100**. In a case where the display unit **114** has a touch panel, the touch panel also serves as the operation unit **115**.

(36) The memory **116** stores various data such as image data subjected to image processing by the image processing unit **113**. The memory **116** also stores programs. The memory **116** is realized by using ROM, RAM, and HDD, for example.

(37) The body system control unit **117** controls the camera system **100** as a whole. The body system control unit **117** is realized by using a CPU, for example.

Example 1

(38) FIG. **5** is a block diagram of a camera system according to this embodiment. The camera system includes the interchangeable lens **200** and the camera body **110**. The interchangeable lens **200** includes the first optical system **201**R and the second optical system **201**L. The interchangeable lens **200** further includes a first driving mechanism **263**R (third adjusting unit) that moves the first optical system **201**R and a second driving mechanism (fourth adjusting unit) **263**L that moves the second optical system **201**L. The interchangeable lens **200** includes a lens type information memory **150**. Here, the lens type information is configuration information of the optical system, and specifically, information including an identifier indicating whether or not the interchangeable lens **200** is a lens for VR imaging.

(39) The camera body **110** includes the image sensor **111**, the operation unit **151**, a parallax calculator (determining unit) **152**, a focus detector (first acquiring unit, second acquiring unit) **153**,

and a driving amount determining unit (control unit) **154**. The parallax calculator **152**, the focus detector **153**, and the driving amount determining unit **154** are included in the body system control unit **117** in this embodiment, but this embodiment is not limited to this example. For example, the lens system control unit **209** may has a configuration having functions equivalent to those of the parallax calculator **152**, the focus detector **153**, and the driving amount determining unit **154**. The parallax calculator **152**, the focus detector **153**, and the driving amount determining unit **154** may be configured as a control apparatus separate from the camera body **110**. The image sensor **111** includes a single image sensor, and two images, an image formed via the first optical system **201**R and an image formed via the second optical system **201**L, are formed on the imaging surface of the image sensor **111**. The operation unit **151** is, for example, a touch panel, a joystick, or the like, and is used by the user to select a focus detection position during autofocusing of the camera system. The parallax calculator **152** calculates a parallax amount between the object image formed via the first optical system **201**R and the object image formed via the second optical system **201**L based on configuration information of the optical system in the lens type information memory **150**. The parallax calculator **152** determines a second focus detection position corresponding to the second optical system **201**L based on the calculated parallax amount and the first focus detection position corresponding to the first optical system **201**R. Here, the first focus detection position and the second focus detection position are imaging positions of the same object. The focus detector **153** acquires a focus detection evaluation value at the focus detection position designated by the operation unit **151** or the parallax calculator **152**. The driving amount determining unit **154** determines driving amounts of the first driving mechanism **263**R and the second driving mechanism **263**L from the focus detection evaluation value acquired by the focus detector **153**.
(40) A description will now be given of a focus detection position determination method by the parallax calculator **152** according to this embodiment. In a camera system for VR imaging which includes an interchangeable lens having a plurality of optical systems and an image pickup apparatus having a single image sensor, in a case where the base length L**1** is set long so that images with a large stereoscopic effect can be captured, it is conceivable to adopt the bending optical system shown in FIG. **1**. At this time, it is necessary to make the distance L**2** between the third optical axes shorter than the base length L**1** and the diameter of the lens mount unit **202**.
(41) The parallax calculator **152** first performs triangulation based on the focal lengths and the base length L**1** of the optical systems stored in the lens type information memory **150**, and the object distance information acquired by the focus detector **153**. Next, the parallax calculator **152** calculates a parallax amount on the imaging surface of the image sensor **111** between the object image formed by the first optical system **201**R and the object image formed by the second optical system **201**L. In order to calculate the parallax with high accuracy, the lens type information memory **150** may store optical information such as projection methods and distortion coefficients of the first optical system **201**R and the second optical system **201**L.
(42) The parallax calculator **152** determines the focus detection position corresponding to the second optical system **201**L using the following equation (1):

$$(X2,Y2)=(X1+Xp+L2+X.\text{sub}.E,Y1+Yp+Y.\text{sub}.E) \qquad (1)$$

where (X1, Y1) are coordinates on the imaging surface of the image sensor **111** of the focus detection position corresponding to the first optical system **201**R. (X2, Y2) are coordinates on the imaging surface of the image sensor **111** of the focus detection position corresponding to the second optical system **201**L. (Xp, Yp) is a parallax amount vector of an object on the imaging surface of the image sensor **111**. (L**2**, **0**) is a distance vector between the third optical axes. (X.sub.E, Y.sub.E) is a shift vector from the ideal value of the distance between the third optical axes due to the mount attachment and detachment operations.
(43) Referring now to FIGS. **6**A and **6**B, a description will be given of a focusing operation of the camera system of this embodiment. FIG. **6**A is an example of a flowchart illustrating the focusing operation of the camera system by the body system control unit **117** according to this embodiment.

(44) In step S**501**, the user operates the operation unit **151** to select a first focus detection position corresponding to the first optical system **201**R.
(45) In step S**502**, the focus detector **153** acquires a focus detection evaluation value (first evaluation value) of the first optical system **201**R at the first focus detection position selected by the user.
(46) In step S**503**, the parallax calculator **152** calculates a parallax amount based on the focal lengths and base length L**1** of optical systems stored in the lens type information memory **150**, and the object distance information acquired by the focus detector **153**. Here, the parallax amount is a parallax amount between the object image formed by the first optical system **201**R and the object image formed by the second optical system **201**L, as described above.
(47) In step S**504**, the parallax calculator **152** determines a second focus detection position corresponding to the second optical system **201**L based on the focus detection position of the first optical system **201**R, the distance L**2** between the third optical axes, and the parallax amount.
(48) In step S**505**, the focus detector **153** acquires a focus detection evaluation value (second evaluation value) of the second optical system **201**L at the determined second focus detection position.
(49) In step S**506**, the driving amount determining unit **154** determines driving amounts of the first driving mechanism **263**R and the second driving mechanism **263**L from the focus detection evaluation value of each optical system.
(50) In step S**507**, the first driving mechanism **263**R and the second driving mechanism **263**L are driven. Thereby, the focusing operations of the first optical system **201**R and the second optical system **201**L are performed.
(51) FIG. **6**B is another example of a flowchart illustrating the focusing operation of the camera system by the body system control unit **117** according to this embodiment.
(52) In step S**511**, the user operates the operation unit **151** to select the first focus detection position corresponding to the first optical system **201**R.
(53) In step S**512**, the focus detector **153** acquires the focus detection evaluation value (first evaluation value) of the first optical system **201**R at the first focus detection position selected by the user.
(54) In step S**513**, the parallax calculator **152** detects feature points in each image formed and acquired by the first optical system **201**R and the second optical system **201**L.
(55) In step S**514**, the parallax calculator **152** matches feature points pointing to the same object in images formed by different optical systems, and uses the result of feature point matching, and determines the second focus detection position corresponding to the second optical system **201**L.
(56) In step S**515**, the focus detector **153** acquires the focus detection evaluation value (second evaluation value) of the second optical system **201**L at the determined second focus detection position.
(57) In step S**516**, the driving amount determining unit **154** determines the driving amounts of the first driving mechanism **263**R and the second driving mechanism **263**L from the focus detection evaluation value of each optical system.
(58) In step S**517**, the first driving mechanism **263**R and the second driving mechanism **263**L are driven. Thereby, the focusing operations of the first optical system **201**R and the second optical system **201**L are performed.
(59) Referring now to FIG. **7**, a description will be given of the operation of attaching the interchangeable lens **200** to the camera body **110**. FIG. **7** is a flowchart illustrating the operation of the camera system by the body system control unit **117** in a case where the interchangeable lens **200** according to this embodiment is attached.
(60) In step S**610**, the body system control unit **117** detects that the interchangeable lens **200** is attached to the camera body **110**.
(61) In step S**620**, the parallax calculator **152** identifies the lens type information stored in the lens

type information memory **150** of the attached interchangeable lens **200**.
(62) In step S**630**, it is determined using the lens type information whether the attached interchangeable lens **200** is a VR imaging lens. In a case where it is determined that the attached interchangeable lens **200** is a VR imaging lens, the processing of step S**640** is executed; otherwise, the processing of step S**670** is executed.
(63) In step S**640**, the parallax calculator **152** selects a focus detection position determination method. In a case where the focus detection position determination method is selected and, for example, the interchangeable lens **200** is a lens for VR180 imaging, the parallax calculator **152** uses equation (1) in this example to determine a focus detection position corresponding to the second optical system **201**L.
(64) In step S**650**, the camera body **110** performs bundle adjustment, for example, and detects an error from the design value and an optical axis shift (misalignment) due to mount attachment and detachment operations.
(65) FIG. **8** is a flowchart illustrating a bundle adjustment method by the body system control unit **117** according to this embodiment.
(66) In step S**651**, the image sensor **111** acquires a pair of images formed by the first optical system **201**R and the second optical system **201**L.
(67) In step S**652**, the parallax calculator **152** detects feature points in the acquired image.
(68) In step S**653**, the parallax calculator **152** matches feature points pointing to the same object in images formed by different optical systems, and extracts n feature point pairs.
(69) In step S**654**, the parallax calculator **152** sets a plurality of error parameters for each optical system. For example, X.sub.ER is a horizontal component of the optical axis shift of the first optical system **201**R due to the mount attachment and detachment operations, Y.sub.ER is a vertical component of the optical axis shift of the first optical system **201**R due to the mount attachment and detachment operations, X.sub.EL is a horizontal component of the optical axis shift of the second optical system **201**L, and Y.sub.EL is a vertical component of the optical axis shift of the second optical system **201**L. At this time, a plurality of parameter sets P are set in which the components X.sub.ER, Y.sub.ER, X.sub.EL, and Y.sub.EL are set slightly differently. That is, the parameter set P is defined as illustrated in equation (2) below:

(70) $$P = \begin{pmatrix} X_{ER} \\ Y_{ER} \\ X_{EL} \\ Y_{EL} \end{pmatrix} \quad (2)$$

(71) In step S**655**, a reprojection error E for each parameter set P is calculated based on the following equation (3):

$E=\|p.\text{sub.i}-f(q.\text{sub.i},P)\|$ (3)

(72) The projection function f is a function for converting the coordinates of the feature point imaged by the first optical system **201**R to the coordinates of the image imaged by the second optical system **201**L based on the focal length and base length of the interchangeable lens **200**.
(73) In step S**656**, the parallax calculator **152** performs optimization calculation by the nonlinear least-squares method illustrated in equation (4) below, and determines one error parameter solution P.sub.ans∈P. A difference between the components X.sub.ER and X.sub.EL of the solution P.sub.ans and a difference between the components Y.sub.ER and Y.sub.EL of the solution P.sub.ans correspond to the shift amount vector (X.sub.E, Y.sub.E) of a distance between the third optical axes from the ideal value due to the mount attachment and detachment operations illustrated in equation (1):

(74) $$P_{ans} = \underset{P}{\operatorname{argmin}} \text{ .Math.}_{i}^{n} \text{ .Math. } p_i - f(q_i, P) \text{ .Math. }^{2} \quad (4)$$

(75) In a case where bundle adjustment is performed, the processing returns to the flow in FIG. **7**. In step S**660** of FIG. **7**, the parallax calculator **152** writes the optical axis shift P.sub.ans due to the mount attachment and detachment operations to the lens type information memory **150**.
(76) In step S**670**, the parallax calculator **152** stops functioning.
(77) The method of calculating the optical axis shift P.sub.ans due to the mount attachment and detachment operations for bundle adjustment is not limited to the method described in this example. For example, a sensor that calculates the optical axis shift P.sub.ans may be mounted. The interchangeable lens **200** may include a calculator for the optical axis shift P.sub.ans.
Example 2
(78) FIG. **9** is a schematic configuration diagram of the camera system **100** according to this example. This example will discuss only configurations different from that of Example 1, and omit a description of common configurations.
(79) The image sensor **111** can perform image-plane phase-difference AF by detecting a focus shift amount and focus shift direction. The interchangeable lens **200** includes a driving mechanism (first adjusting unit) **261** configured to move the lens top base **300** and a driving mechanism (second adjusting unit) **262** that is provided on the lens top base **300** and moves the first optical system **201**R. The driving mechanism **261** can move the first optical system **201**R and the second optical system **201**L in a direction orthogonal to the imaging surface of the image sensor **111** by moving the lens top base **300**.
(80) The second optical system **201**L is fixed to the lens top base **300**. The first optical system **201**R is supported by the lens top base **300** so as to be movable in the direction orthogonal to the imaging surface of the image sensor **111** relative to the lens top base **300** by the driving mechanism **262**. Thereby, the first optical system **201**R and the second optical system **201**L can move relative to each other in the direction orthogonal to the imaging surface of the image sensor **111**. In this example, each of the first optical system **201**R and the second optical system **201**L includes a lens unit in which an imaging optical system is integrated, and focusing can be performed by extending the entire optical system. This example entirely extends the two optical systems, and can reduce a characteristic difference between the two optical systems. In this example, the first optical system **201**R is a first focus lens optical system, and the second optical system **201**L is a second focus lens optical system. The second optical system **201**L may be the first focus lens optical system, and the first optical system **201**R may be the second focus lens optical system.
(81) The image sensor **111** is installed so that its imaging surface is parallel to the lens mount unit **202**. However, it is difficult to make the imaging surface perfectly parallel to the lens mount unit **202** due to manufacturing errors, and the image sensor **111** is actually fixed with its imaging surface slightly tilted relative to the lens mount unit **202**. FIG. **10** is a schematic configuration diagram illustrating the tilted image sensor **111**. The manufacturing process adjusts the interchangeable lens **200** so that a distance between the imaging position of the first optical system **201**R and the imaging position of the second optical system **201**L from the lens mount unit **202**, that is, a difference between the so-called flange back distances becomes 0. However, due to the tilt of the image sensor **111**, the two optical systems do not always have the best in-focus position. Accordingly, this example configures the two optical systems movable in the direction orthogonal to the imaging surface by the driving mechanism **262**, and thereby adjusts the focal positions of the two optical systems.
(82) Referring now to FIGS. **11**A, **11**B, and **11**C, a description will be given of the focusing operation procedure according to this example. FIGS. **11**A, **11**B, and **11**C illustrate the focusing operation procedure according to this example.
(83) FIG. **11**A illustrates the camera system **100** before focusing. Assume that a focus difference (left-right focus difference) H1 between the two optical systems can be simply expressed by a difference between an R1 surface of the first lens **211**R disposed in the first optical system **201**R and an R1 surface of the first lens **211**L disposed in the second optical system **201**L. A focus shift

amount is defined as a difference between the R1 surface of the first lens **211**R(L) and a line **20**OA provided on the interchangeable lens **200**, and a focus shift amount of the first optical system **201**R can be simply represented by H2R, a focus shift amount of the second optical system **201**L can be simply represented by H2L. The in-focus state is achieved in a case where the R1 surface of the first lens unit **211**R(L) and the line **20**OA provided on the interchangeable lens **200** overlap each other, that is, in a case where the focus shift amounts H2R and H2L become zero.
(84) In a case where the user presses the release button, the focus detector **153** acquires two evaluation values (moving amount and moving direction) corresponding to the two optical systems, and determines the focus shift amount. In a case where the focus detection position of one of the two optical systems is determined based on the user's input, the focus detection position of the other of the two optical systems is determined according to the focus detection position determination method selected by the parallax calculator **152**. Based on the evaluation values obtained at the determined focus detection positions, it is confirmed whether the difference between the two evaluation values (left-right focus difference H1) is within the permissible range (predetermined value).
(85) In this example, since the left-right focus difference H1 is not within the permissible range, the first optical system **201**R is moved using the driving mechanism **262** so that the left-right focus difference H1 becomes within the permissible range. FIG. **11**B illustrates the R1 surfaces of the first lenses **211**R and **211**L on the same plane, and illustrates that the left-right focus difference H1 is controlled to be within the permissible range. Thereafter, as illustrated in FIG. **11**C, the driving mechanism **261** simultaneously moves the two optical systems so that the focus shift amounts H2R and H2L are within the permissible range.
(86) In this example, the driving mechanism **261** is driven after the driving mechanism **262** is driven.
(87) As illustrated in FIG. **12**A, first, the driving mechanism **261** may be driven so that the focus shift amount H2L of the second optical system **201**L is within the permissible range. Thereafter, as illustrated in FIG. **12**B, the driving mechanism **262** may be driven so that the focus shift amount H2R of the first optical system **201**R is within the permissible range.
(88) Referring now to FIGS. **13**A and **13**B, a description will be given of a focusing operation procedure in a case where the focus shift amount H2L of the second optical system **201**L is within the permissible range and the focus shift amount H2R of the first optical system **201**R and the left-right focus difference H1 are substantially equal.
(89) FIG. **13**A illustrates the camera system **100** before focusing. In a case where the user presses the release button, the focus detector **153** acquires two evaluation values (moving amount and moving direction) corresponding to the two optical systems, and determines the focus shift amounts. Next, it is confirmed whether a difference between the two evaluation values (left-right focus difference H1) is within the permissible range.
(90) In this example, the left-right focus difference H1 is not within the permissible range, the first optical system **201**R is moved using the driving mechanism **262** so that the left-right focus difference H1 is within the permissible range. The focus shift amount H2L of the second optical system **201**L is originally within the permissible range, and as illustrated in FIG. **13**B, the focus shift amount H2R of the first optical system **201**R becomes also within the permissible range and the in-focus states are acquired.
(91) Referring now to **14**A and **14**B, a description will be given of a focusing operation procedure in a case where the left-right focus difference H1 is within the permissible range and the focus shift amount H2R of the first optical system **201**R and the focus shift amount H2L of the second optical system **201**L are equal.
(92) FIG. **14**A illustrates the camera system **100** before focusing. In a case where the user presses the release button, the focus detector **153** acquires two evaluation values (moving amount and moving direction) corresponding to the two optical systems, and determines the focus shift

amounts. Next, it is confirmed whether a difference between the two evaluation values (left-right focus difference H1) is within the permissible range. In this example, since the left-right focus difference H1 is within the permissible range, as illustrated in FIG. **14**B, the driving mechanism **261** simultaneously moves the two optical systems so that the focus shift amounts of the two optical systems are within the permissible ranges.

Example 3

(93) Another example will be described. The basic mechanical configuration or the like are the same as that of FIG. **14** according to Example 2 described above, and this example will discuss only the configuration different from that of Example 2, and will omit a description of the common configuration.

(94) In order to simplify the operation, in this example, assume that focus shift amounts of the two optical systems are adjusted in advance by any method so that they are substantially the same. In that case, it is unnecessary to determine a second focus detection position and to acquire a corresponding evaluation value (moving amount and moving direction) that is the focus shift amount H2L of the second optical system, and thus the operation flow of the camera system **100** becomes simple.

(95) That is, before imaging, adjustment is performed through manual operation so that the left-right focus difference H1 between the first optical system **201**R and the second optical system **201**L is within the permissible range. This adjustment can perform subsequent imaging without worrying about the left-right focus difference H1 between the first optical system **201**R and the second optical system **201**L. Sending an operation signal from the outside to the driving mechanism **262** to move only the first optical system **201**R can adjust the focus difference H1 between the two optical systems **201**R and **201**L to be within the permissible range.

(96) The driving mechanism **262** includes a stepping motor, gears, etc., and can be operated by an operation signal generated by a detector that detects the operation amount in a case where the user operates an operation ring of the camera, etc. Thereafter, an imaging operation is performed.

(97) In a case where the user presses the release button, the focus detector **153** acquires only one evaluation value (moving amount and moving direction) corresponding to the first optical system **201**R and the focus shift amount H2R. Based on the focus shift amount H2R, the driving mechanism **261** is driven so that the focus shift amount H2R of the first optical system is within the permissible range. By driving the driving mechanism **261**, the first optical system **201**R and the second optical system **201**L are moved in the same direction by the same moving amount, and the two optical systems can become in in-focus states simultaneously.

Example 4

(98) Another example will be described. The basic mechanical configuration or the like is the same as that of Example 2 or 3 described above, so this example will discuss only the configuration different from that of Example 2 or 3, and will omit a description of the common configuration.

(99) FIG. **21**A illustrates another configuration that can previously adjusted the left-right focus difference H1 between the first optical system **201**R and the second optical system **201**L to be within the permissible range through manual operation. An adjusting mechanism **264** can move the first optical system **201**R in the optical axis direction. The adjusting mechanism **264** includes an unillustrated eccentric roller and the like, and is configured to be able to rotate the eccentric roller from the outer circumferential portion of the interchangeable lens **200** with a hexagon wrench or the like. By this operation, only the first optical system **201**R can be moved, and the focus difference H1 between the two optical systems **201**R and **201**L can be adjusted to be within the permissible range. Thereafter, an imaging operation is performed.

(100) In a case where the user presses the release button, the focus detector **153** acquires only one evaluation value (moving amount and moving direction) corresponding to the first optical system **201**R and the focus shift amount H2R. Based on the focus shift amount H2R, the driving mechanism **261** is driven so that the focus shift amount H2R of the first optical system is within the

permissible range (FIG. **21**B). By driving the driving mechanism **261**, the first optical system **201**R and the second optical system **201**L are moved in the same direction by the same moving amount, and the two optical systems can become in in-focus states simultaneously.

Example 5

(101) FIG. **15** is a schematic configuration diagram of the camera system **100** according to this example. This example will discuss only configurations different from those of Examples 1 to 4, and will omit a description of common configurations.

(102) The interchangeable lens **200** includes a first driving mechanism **263**R and a second driving mechanism **263**L. The first driving mechanism **263**R and the second driving mechanism **263**L are attached to the lens top base **300**. That is, the first driving mechanism **263**R and the second driving mechanism **263**L are disposed on the same member. The first driving mechanism **263**R moves the first optical system **201**R relative to the lens top base **300** in the direction orthogonal to the imaging surface of the image sensor **111**. The second driving mechanism **263**L moves the second optical system **201**L relative to the lens top base **300** in the direction orthogonal to the imaging surface of the image sensor **111**. Thereby, the first optical system **201**R and the second optical system **201**L can move relative to each other in the direction orthogonal to the imaging surface of the image sensor **111**.

(103) In this example, each of the first optical system **201**R and the second optical system **201**L includes a lens unit in which an imaging optical system is integrated, and focusing can be performed by extending the entire optical system. This example performs focusing by extending the entire optical system, but may perform focusing by partially extending the optical system or by an inner focus type configuration.

(104) Referring now to FIGS. **16**A and **16**B, a description will be given of a focusing operation procedure according to this example. FIGS. **16**A and **16**B illustrate the focusing operation procedure according to this example.

(105) In a case where the user presses the release button, the focus detector **153** acquires two evaluation values (moving amount and moving direction) corresponding to the two optical systems, and determines the focus shift amounts. Next, it is determined whether the difference between the two evaluation values (left-right focus difference H1) is within the permissible range (predetermined range).

(106) Next, as illustrated in FIG. **16**A, the second driving mechanism **263**L moves the second optical system **201**L so that the focus shift amount H2L is within the permissible range. Thereafter, as illustrated in FIG. **16**B, the first driving mechanism **263**R moves the first optical system **201**R so that the focus shift amount H2R is within the permissible range.

(107) This example drives the first driving mechanism **263**R, after driving the second driving mechanism **263**L. However, this embodiment may drive the second driving mechanism **263**L after driving the first driving mechanism **263**R. Without any problems of power limitation, the first driving mechanism **263**R and the second driving mechanism **263**L may be driven simultaneously.

(108) Another example will be described. In order to simplify the operation, assume that focus shift amounts of the two optical systems have been previously adjusted by any method so that they are approximately the same. In that case, it is unnecessary to determine the second focus detection position and to acquire the corresponding evaluation value (moving amount and moving direction) that is the focus shift amount H2L of the second optical system, and the operation flow of the camera system **100** becomes simple.

(109) The method for the user to adjust the focus difference H1 between the two optical systems **201**R and **201**L to be within the permissible range is as described above, and can be selected from various configurations. Thereafter, an imaging operation is performed.

(110) In a case where the user presses the release button, the focus detector **153** acquires only one evaluation value (moving amount and moving direction) corresponding to the first optical system **201**R and the focus shift amount H2R. Based on the focus shift amount H2R, the driving

mechanism **263**R is driven so that the focus shift amount H2R of the first optical system is within the permissible range. Thereafter, by driving the driving mechanism **263**L, the second optical system **201**L is moved in the same direction by the same moving amount. Without any problems of power limitation, the first driving mechanism **263**R and the second driving mechanism **263**L may be simultaneously driven. Since the evaluation value (moving amount and moving direction), which is the focus shift amount H2L of the second optical system, is not used, the operation flow becomes simple.

Example 6

(111) FIG. **17** is a schematic configuration diagram of a camera system **100** according to this example. The basic configuration of the camera system **100** according to this example is the same as that of each of the above examples. This example will discuss only the configuration different from that of each of the above examples, and will omit a description of the common configuration.

(112) The interchangeable lens **200** includes a position detection sensor (first detector) **271**R configured to detect the position of the first optical system **201**R and a position detection sensor (second detector) **271**L configured to detect the position of the second optical system **201**L. The interchangeable lens **200** further includes a guide portion (first guide portion) **272**R configured to guide the first optical system **201**R and a guide portion (second guide portion) **272**L configured to guide the first optical system **201**R.

(113) The first driving mechanism **263**R and the second driving mechanism **263**L are mounted on the lens top base **300** so as to be point-symmetrical with respect to the center of the lens mount unit **202** (the center of the interchangeable lens **200**) in a case where the interchangeable lens **200** is viewed from the object side. The first driving mechanism **263**R and the second driving mechanism **263**L are disposed in opposite directions.

(114) The position detection sensors **271**R and **271**L are also attached to the lens top base **300**. The attachment to the lens top base **300** can detect the position detection sensors **271**R and **271**L with high accuracy. The guide portions **272**R and **272**L are also attached to the lens top base **300**, respectively, and guide the first optical system **201**R and the second optical system **201**L.

Example 7

(115) FIG. **18** illustrates the interchangeable lens **200** according to this example viewed from the lens mount unit **202** side. This example will discuss only configurations different from those of Examples 1 to 6, and will omit a description of common configurations.

(116) In FIG. **18**, a contact **281** with the camera body **110** is disposed below the first optical system **201**R and the second optical system **201**L, and allows the camera body **110** and the interchangeable lens **200** to electrically communicate with each other. In this example, signals are exchanged between the camera body **110** and the interchangeable lens **200** through the contact **281**.

(117) FIG. **19** illustrates the interchangeable lens **200** viewed from the object side. FIG. **19** omits unnecessary portions so that the location of the driving mechanism **261** can be viewed. Since a flexible printed circuit (FPC) board **282** is connected to the contact **281**, the contact **281** and the FPC board **282** are disposed substantially in the same phase. In order to avoid electrical interference, the driving mechanism **261** and the contact **281** are disposed at positions different in phase by 180 degrees with respect to the center of the optical axis. The 180-degree difference includes not only a strict 180-degree difference, but also a substantial 180-degree difference (approximate 180-degree difference).

(118) FIG. **20** is a top view of the interchangeable lens **200** viewed from the object side by adding the first optical system **201**R and the second optical system **201**L to the configuration of FIG. **19**. The driving mechanism **261** is disposed outside the projection surface areas of the two optical systems. Disposing the driving mechanism **261** avoiding the space where the two optical systems are disposed, the space can be effectively utilized.

(119) This embodiment can provide a control apparatus that can properly perform focusing of a plurality of optical systems.

OTHER EMBODIMENTS

(120) Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer-executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer-executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer-executable instructions. The computer-executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read-only memory (ROM), a storage of distributed computing systems, an optical disc (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

(121) While the disclosure has been described with reference to embodiments, it is to be understood that the disclosure is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(122) This application claims the benefit of Japanese Patent Applications Nos. 2022-157404, filed on Sep. 30, 2022, and 2023-090520, filed on May 31, 2023, each of which is hereby incorporated by reference herein in their entirety.

Claims

1. A control apparatus for a camera system that includes a lens apparatus including a first optical system and a second optical system and configured to move the first optical system and the second optical system relative to each other, and an image pickup apparatus including an image sensor and attachable to and detachable from the lens apparatus, the control apparatus comprising: a memory storing instructions; and a processor configured to execute the instructions to: acquire a first evaluation value of the first optical system at a first focus detection position corresponding to the first optical system, determine a second focus detection position corresponding to the second optical system based on an first object image formed by the first optical system and a second object image formed by the second optical system, acquire a second evaluation value of the second optical system at the second focus detection position, and move the first optical system and the second optical system based on the first evaluation value and the second evaluation value.

2. The control apparatus according to claim 1, wherein the first focus detection position and the second focus detection position are imaging positions of a same object.

3. The control apparatus according to claim 1, wherein the processor is configured to determine the second focus detection position using the first focus detection position, a parallax amount between the first object image and the second object image, a distance between optical axes of the first optical system and the second optical system set closest to an imaging surface of the image sensor, and an optical axis shift due to mount attachment and detachment operations.

4. The control apparatus according to claim 1, wherein the processor is configured to determine the second focus detection position using feature point matching between the first object image and the second object image.

5. The control apparatus according to claim 1, wherein the processor is configured to acquire from the lens apparatus, information about the lens apparatus including information identifying whether the lens apparatus is a VR imaging lens.
6. The control apparatus according to claim 5, wherein the lens apparatus is determined to be the VR imaging lens, the processor acquires an error from a design value of the lens apparatus.
7. The control apparatus according to claim 6, wherein the error is an optical axis shift due to mount attachment and detachment operations of the lens apparatus.
8. The control apparatus according to claim 5, wherein the processor is configured to determine the second focus detection position in a case where the processor determines that the lens apparatus is the VR imaging lens.
9. The control apparatus according to claim 1, wherein the lens apparatus includes: a first adjusting unit configured to simultaneously move the first optical system and the second optical system; and a second adjusting unit configured to move the first optical system, wherein the processor is configured to drive the first adjusting unit and the second adjusting unit based on a difference between the first evaluation value and the second evaluation value.
10. The control apparatus according to claim 9, wherein the processor is configured to drive one of the first adjusting unit and the second adjusting unit in a case where the difference is larger than a predetermined value, and then to drive the other of the first adjusting unit and the second adjusting unit.
11. The control apparatus according to claim 9, wherein the processor is configured to drive only the first adjusting unit in a case where the difference is smaller than a predetermined value.
12. The control apparatus according to claim 1, wherein the lens apparatus includes: a third adjusting unit configured to move the first optical system; and a fourth adjusting unit configured to move the second optical system, wherein the processor is configured to drive the third adjusting unit based on the first evaluation value and to drive the fourth adjusting unit based on the second evaluation value.
13. A lens apparatus attachable to and detachable from an image pickup apparatus that includes an image sensor, the lens apparatus comprising: a first optical system; a second optical system; and the control apparatus according to claim 1.
14. The lens apparatus of claim 13, further comprising: a first adjusting unit configured to simultaneously move the first optical system and the second optical system; and a second adjusting unit configured to move the first optical system.
15. The lens apparatus according to claim 14, wherein in a case where the lens apparatus is viewed from an object side, the first adjusting unit is disposed at a position different in phase from a contact with the image pickup apparatus by 180 degrees with respect to a center of the lens apparatus.
16. The lens apparatus according to claim 14, wherein in a case where the lens apparatus is viewed from an object side, the first adjusting unit is disposed outside projection surface areas of the first optical system and the second optical system.
17. The lens apparatus of claim 13, further comprising: a third adjusting unit configured to move the first optical system; and a fourth adjusting unit configured to move the second optical system.
18. The lens apparatus according to claim 17, wherein the third adjusting unit and the fourth adjusting unit are disposed on a same member.
19. The lens apparatus according to claim 17, wherein the third adjusting unit and the fourth adjusting unit are disposed point-symmetrically with respect to a center of the lens apparatus in a case where the lens apparatus is viewed from an object side.
20. The lens apparatus according to claim 17, further comprising: a first detector configured to detect a position of the first optical system; and a second detector configured to detect a position of the second optical system, wherein the first detector and the second detector are disposed on a same member.

21. The lens apparatus according to claim 17, further comprising: a first guide portion configured to guide the first optical system; and a second guide portion configured to guide the second optical system, wherein the first guide portion and the second guide portion are disposed on a same member.
22. An image pickup apparatus attachable to and detachable from a lens apparatus that includes a first optical system and a second optical system, and is configured to move the first optical system and the second optical system relative to each other, the image pickup apparatus comprising: an image sensor; and the control apparatus according to claim 1.
23. A camera system comprising: a lens apparatus including a first optical system and a second optical system, and configured to move the first optical system and the second optical system relative to each other; an image pickup apparatus including a single image sensor and attachable to and detachable from the lens apparatus; and the control apparatus according to claim 1.
24. A control method for a camera system that includes a lens apparatus including a first optical system and a second optical system and is configured to move the first optical system and the second optical system relative to each other, and an image pickup apparatus that includes an image sensor and is attachable to and detachable from the lens apparatus, the control method comprising the steps of: acquiring a first evaluation value of the first optical system at a first focus detection position corresponding to the first optical system, determining a second focus detection position corresponding to the second optical system based on an first object image formed by the first optical system and a second object image formed by the second optical system, acquiring a second evaluation value of the second optical system at the second focus detection position, and moving the first optical system and the second optical system based on the first evaluation value and the second evaluation value.
25. A control apparatus for a camera system that includes a lens apparatus including a first optical system and a second optical system and is configured to move the first optical system and the second optical system relative to each other, and an image pickup apparatus that includes an image sensor and is attachable to and detachable from the lens apparatus, the control apparatus comprising: a memory storing instructions; and a processor configured to execute the instructions to: acquire a first evaluation value of the first optical system at a first focus detection position corresponding to the first optical system, and move the first optical system and the second optical system based on the first evaluation value.
26. The control apparatus according to claim 25, wherein the lens apparatus includes: a first adjusting unit configured to simultaneously move the first optical system and the second optical system; and a second adjusting unit configured to move the first optical system, wherein the processor drives the first adjusting unit based on the first evaluation value.
27. The control apparatus according to claim 25, wherein a second adjusting unit configured to move the first optical system adjusts a focus difference between the first optical system and the second optical system to be within a permissible range by sending an operation signal to the second adjusting unit through manual operation.
28. The control apparatus according to claim 25, wherein a second adjusting unit configured to move the first optical system adjusts a focus difference between the first optical system and the second optical system to be within a permissible range by operating the second adjusting unit through manual operation.
29. The control apparatus according to claim 25, wherein the lens apparatus includes: a third adjusting unit configured to move the first optical system; and a fourth adjusting unit configured to move the second optical system, wherein the processor drives the third adjusting unit and the fourth adjusting unit based on the first evaluation value.
30. A lens apparatus attachable to and detachable from an image pickup apparatus that includes an image sensor, the lens apparatus comprising: a first optical system; a second optical system; and the control apparatus according to claim 25.

31. The lens apparatus according to claim 30, further comprising: a first adjusting unit configured to simultaneously move the first optical system and the second optical system; and a second adjusting unit configured to move the first optical system.
32. The lens apparatus of claim 30, further comprising: a third adjusting unit configured to move the first optical system; and a fourth adjusting unit configured to move the second optical system.
33. An image pickup apparatus attachable to and detachable from a lens apparatus that includes a first optical system and a second optical system and is configured to move the first optical system and the second optical system relative to each other, the image pickup apparatus comprising: an image sensor; and the control apparatus according to claim 25.
34. A camera system comprising: a lens apparatus including a first optical system and a second optical system, and configured to move the first optical system and the second optical system relative to each other; an image pickup apparatus including a single image sensor and attachable to and detachable from the lens apparatus; and the control apparatus according to claim 25.
35. A control method for a camera system that includes a lens apparatus that includes a first optical system and a second optical system and is configured to move the first optical system and the second optical system relative to each other, and an image pickup apparatus that includes an image sensor and is attachable to and detachable from the lens apparatus, the control method comprising the steps of: acquiring a first evaluation value of the first optical system at a first focus detection position corresponding to the first optical system; and moving the first optical system and the second optical system based on the first evaluation value.